

Chapter 01

Logging

Log Ingestion

Process of collecting logs

Gather data from
multiple systems

Centralize log information

Supports analysis workflows

Log Format Variability

Formats differ across systems

Fields vary in number and order

Date formats are inconsistent

Complicates data processing

Log Type Differences

Different log categories exist
Data included varies by system
Operating systems differ
Device types influence content

Securing Log Data

Logs must be protected

Prevent unauthorized changes

Preserve data integrity

Safeguard sensitive information

Integrity and Encryption

Use hashing for integrity

Detect log tampering

Apply encryption for protection

Secure stored data

Time Synchronization

Logs must align in time

Ensure consistent timestamps

Critical for correlation

Supports accurate analysis

NTP and Time Servers

Use NTP for synchronization

Central time sources

Keep systems aligned

Prevent time discrepancies

Retention Considerations

Storage is limited resource

Requires defined policies

Determine retention period

Manage long-term data use

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Basic Terms and Concepts

The CIA Triad

Foundation of cybersecurity

- Confidentiality – Protect secrets
- Integrity – Data stays unchanged
- Availability – Access when needed

Triple A (AAA)

Three pillars of access control

- Authentication – Prove identity
- Authorization – Access what's allowed
- Accounting – Track all activity

Vulnerabilities, Threats, and Risk

Distinct terms, specific meanings

- Vulnerability – System weakness
- Threat – Weakness plus actor
- Risk – Likelihood meets impact

Malware

“Malicious” + “Software”

- Software with harmful intent
- Viruses, worms, trojans, etc.
- If malicious – it’s malware

Zero Day Attacks

Known only to the attacker

- Undisclosed vulnerability
- No patches available
- Extremely dangerous exploits

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File Formats

File Formats Overview

Many file formats exist

Four Key types for exam

XML, JSON, YAML, and EVTX

Used in cybersecurity data

XML Structure and Characteristics

Extensible Markup
Language (XML)

Based on HTML Structure

Uses opening and closing tags

Fields must match tag names

```
<user>  
  <name>Alice</name>  
  <role>Administrator</role>  
</user>
```

XML Limitations and Usage

Highly verbose format

Can be cumbersome to manage

Requires pattern matching

Widely adopted historically

```
<user>  
  <name>Alice</name>  
  <role>Administrator</role>  
</user>
```

JSON Structure and Usage

JAVAScript Object
Notation (JSON)

Less verbose than XML

Uses key:value pairs

Fields separated by colon

```
{  
  "name": "Alice",  
  "role": "Administrator"  
}
```

JSON Adoption and Parsing

Works well with JavaScript

Supported by many languages

Easier to parse than XML

Common in
automation workflows

```
{  
  "name": "Alice",  
  "role": "Administrator"  
}
```

YAML Structure and Format

Yet Another Markup
Language (YAML)

Less verbose than JSON

Minimal syntax structure

Uses dash-based formatting

```
name: Alice  
role: Administrator
```

EVTX Event Log Format

Windows event log format

Successor to EVT format

XML-based structure

Used for log data

Event ID: 4624

Source: Security

Message: *An account was succesfully logged on*